

Modelling Particle Production in Analog (3+1) Inflationary Cosmologies Using High-Performance Computing

Nathaniel Chapman

Department of Computer Science

Department of Physics

Central Washington University

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Introduction - Analog Universes

① Inflation created particles

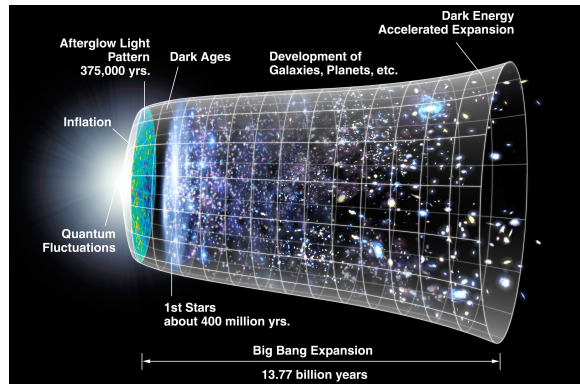


Figure: History of the universe highlighting period of rapid expansion (Source)

Introduction - Analog Universes

- 1 Inflation created particles
- 2 Cannot directly observe these particles

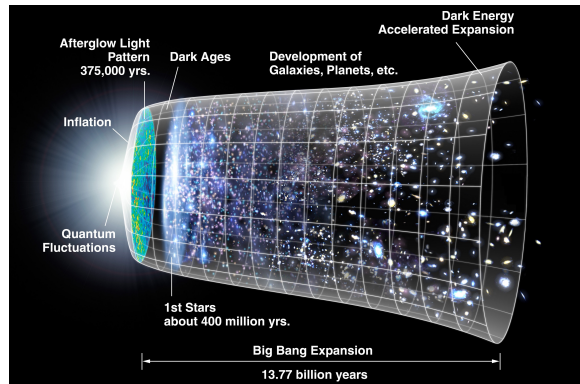


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- 3 Theory suggests ultra-cold quantum gases behave similarly

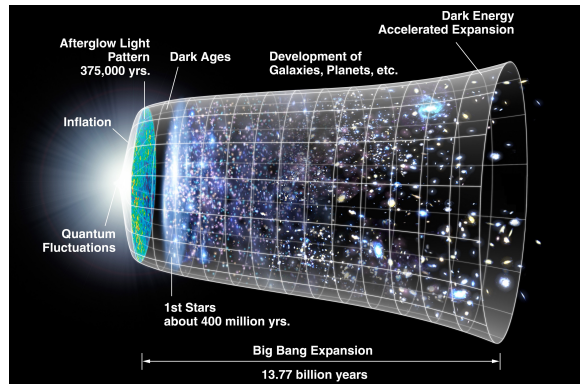


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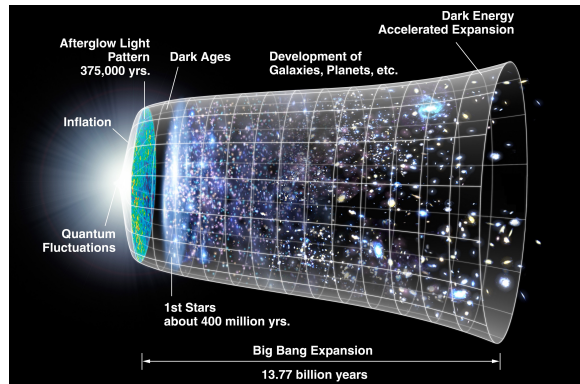


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- 5 Lab based system = “analog universe”

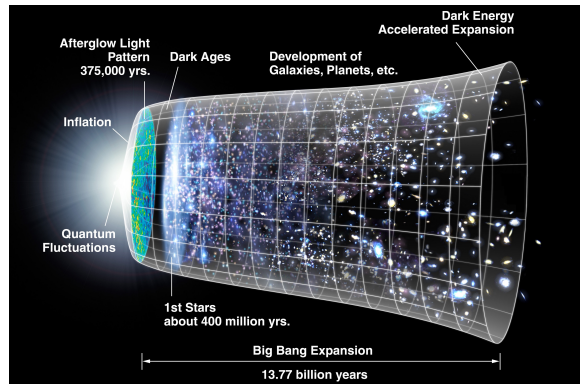


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Introduction - The Story So Far

- Many investigations into analog universes



Figure: Subsets of increasingly specific literature

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- Few into exact analogs
- Fewer into computational simulation
- Even fewer using high-performance computing



Figure: Subsets of increasingly specific literature

Methods - Physical

Cosmological

① Spacetime

- increase in spatial-volume (inflation)
- constant speed of light

② Observer dependent quantum field excitations

③ Spontaneous particle production

Inflation determines particle momentum

Condensed Matter

① Bose-Einstein condensate

- fixed volume
- decreasing speed of sound

② Oscillatory phase perturbations

③

Phase perturbations are damped oscillators

- The Field Equation
- $k < k_c$
- $0 \leq t \leq t_f$
- initial conditions

Methods - Computational

- Symplectic integrator e.g. leapfrog to minimize energy drift
- Parareal for parallel-in-time i.e. predictor-corrector
- GPU
- Distributed on CWU's supercomputer with 12 P100 GPUs

Methods - The Field Equation

- For wave-like perturbations One equation describes both ultra-cold quantum gases and inflation particles
- One equation for each phononic k
- Universe expansion starts and stops at finite times

The Field Equation

$$\partial_t^2 \tilde{\theta}_{\mathbf{k}} - \frac{\dot{a}}{a} \partial_t \tilde{\theta}_{\mathbf{k}} - a c_0^2 k^2 \tilde{\theta}_{\mathbf{k}} = 0$$

$$\tilde{\theta}_{\mathbf{k}}(0) = \tilde{\theta}_{\mathbf{k}0} \quad \partial_\eta \tilde{\theta}_{\mathbf{k}}(0) = -U_0 n_{\mathbf{k}0} / \hbar$$

Methods - Solving The Field Equation

- Need to try to conserve energy

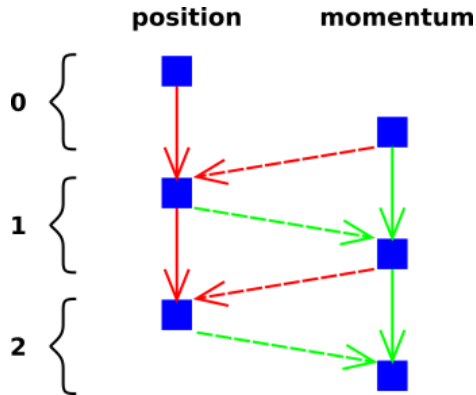


Figure: Symplectic integrator example: Velocity Verlet (Source)

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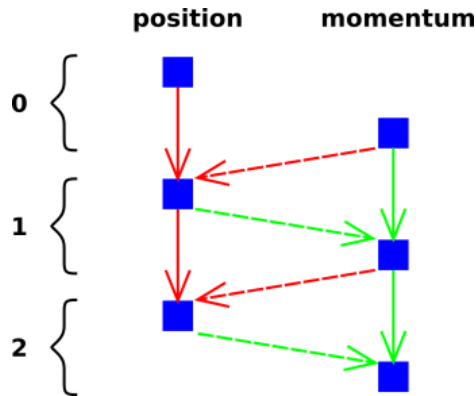


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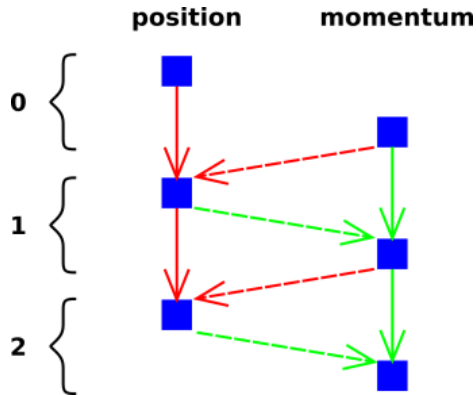


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Methods - Solving The Field Equation

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- Use new momentum to calculate position
- More complex but more stable

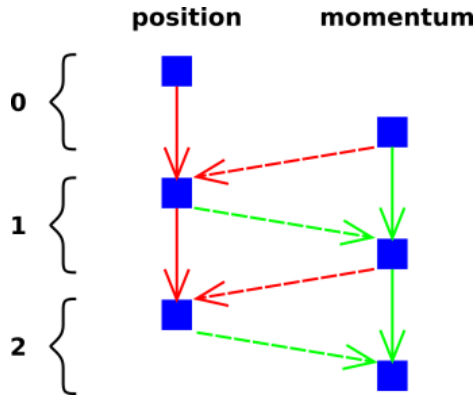


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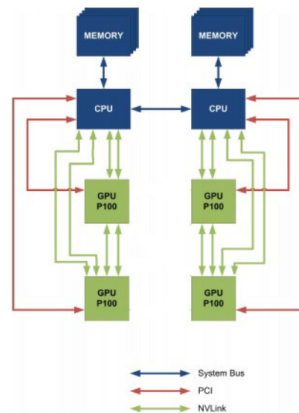


Figure: CPU to GPU and GPU to GPU interconnect using NVLink

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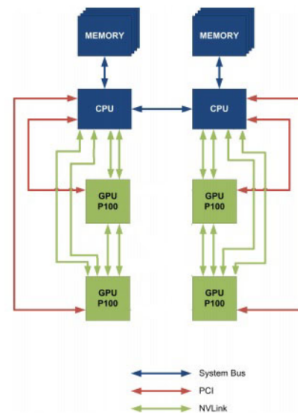


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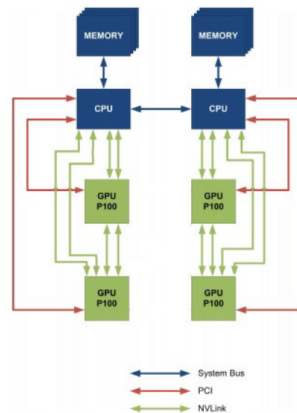


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Methods - High-Performance Computing

- Equation for each $k < k_c$
- Many equations to solve
- Solve at the same time with multiple GPUs
- Use CWU's "Turing" supercomputer
- 4x4 NVIDIA P100 GPUs
 - Each 83% performance of modern RTX 3060

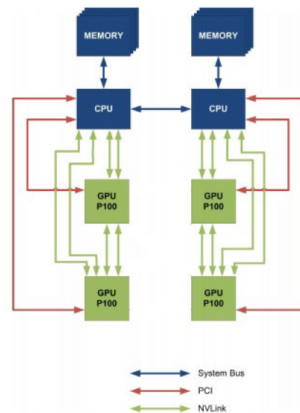


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Analytic Results - Simple Expansion

- Simple expansion: $a(t) = e^{-t/t_s}$

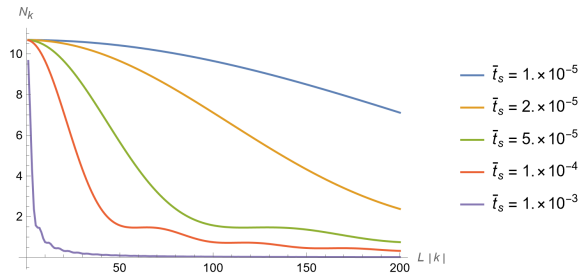


Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

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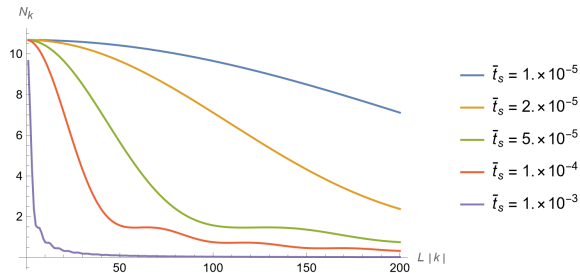


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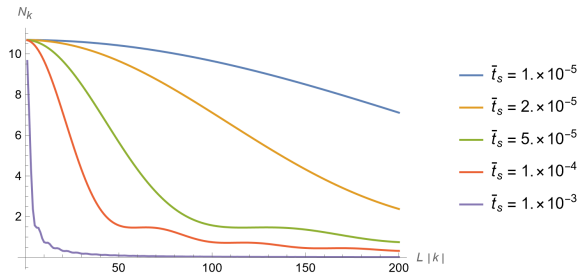


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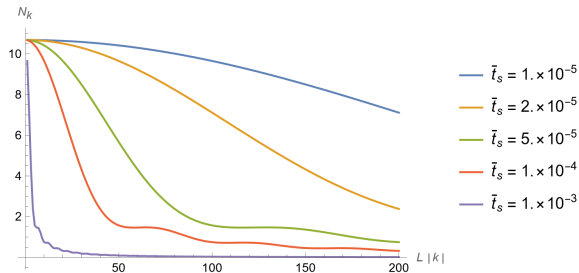


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- Population drops off quickly with k

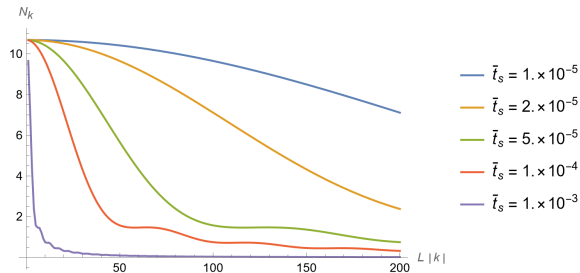


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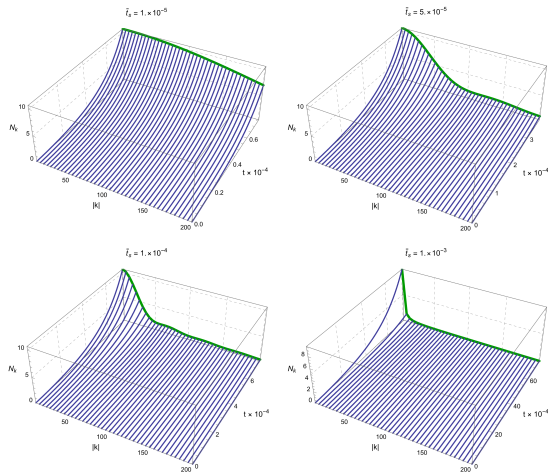


Figure: Momentum distribution throughout the expansion

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- Expect the distribution of momenta to depend on the length of the expansion

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- Theory suggests inflation particles behave the same as ultra-cold quantum gases in the lab
- There are many equations that need to be solved with special methods to conserve energy
- Use high-performance techniques and hardware for reasonable execution time
- Expect the distribution of momenta to depend on the length of the expansion
- Expect the population to be high for low k and drop off quickly

Acknowledgements

- Dr. Andy Piacsek (CWU, Physics)
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Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): github.com/nondairyneutrino/thesis
- My website: nathanwchapman.com

